

References

1. A. Pedretti, A. Mazzolari, S. Gervasoni, L. Fumagalli, G. Vistoli "THE VEGA SUITE OF PROGRAMS: AN VERSATILE PLATFORM FOR CHEMINFORMATICS AND DRUG DESIGN PROJECTS" *Bioinformatics*, Vol. 37(8) 1174-1175 (2021). DOI: 10.1093/bioinformatics/btaa774
2. Abbasi, M., Mansourian, M., Arefi Oskouie, A., Taheri, S., & Mahnam, K. (2024). Insilico study of MM/GBSA binding free energy and molecular dynamics simulation of designed remdesivir derivatives as potential SARSCoV2 main protease inhibitors. *Research in Pharmaceutical Sciences*, 19(1), 29–41. <https://doi.org/10.4103/1735-5362.394818>
3. Abou Hajal, A., & Al Meslamani, A. Z. (2024). Overcoming barriers to machine learning applications in toxicity prediction. *Expert Opinion on Drug Metabolism & Toxicology*, 20(7), 549–553. <https://doi.org/10.1080/17425255.2023.2294939>
4. Acharya, et al. (2023.) Molecular medicinal insights into scaffold hopping-based drug discovery success. <https://www.sciencedirect.com/science/article/abs/pii/S1359644623003616?via%3Dihub>
5. Adelusi, T. I., Oyedele, A. K., Boyenle, I. D., Ogunlana, A. T., Adeyemi, R. O., Ukachi, C. D., Idris, M. O., Olaoba, O. T., Adedotun, I. O., Kolawole, O. E., Xiaoxing, Y., & Abdul-Hammed, M. (2022). Molecular modeling in drug discovery. *Informatics in Medicine Unlocked*, 29, 100880. <https://doi.org/10.1016/j.imu.2022.100880>
6. Aghajani, J., Farnia, P., Farnia, P., Ghanavi, J., & Velayati, A. A. (2022, January 1). Molecular Dynamic Simulations and Molecular Docking as a Potential Way for Designed New Inhibitor Drug without Resistance. <https://pmc.ncbi.nlm.nih.gov/articles/PMC9571241/> Agu, C., et al. (2021). *Bioinformatics*, <https://doi.org/10.1093/bioinformatics/btaa105>
7. Ahmed, Maqsood, et al. "Comprehensive Phytochemical Profiling, Biological Activities, and Molecular Docking Studies of *Pleurosporum Candollei*: An Insight Into Potential for Natural Products Development." *Molecules*, vol. 27, no. 13, June 2022, p. 4113. <https://doi.org/10.3390/molecules27134113>.
8. AlKarboush, et al. (2025). Fragment-based drug discovery: A graphical review. <https://www.sciencedirect.com/science/article/pii/S2590257125000215>
9. Al-Khafaji, K., Al-Duhaidahawi, L., Taskin Tok, T., Al-Shuhaib, M. B. S., Al-Shammari, A. M., & Jasim, S. A. (2023). Identification of potential SARS-CoV-2 main protease inhibitors from natural products using molecular docking, molecular dynamics simulation, and MM-GBSA calculations. *Pharmaceutics*, 15(1), 1–21. <https://doi.org/10.3390/pharmaceutics15010011>

10. Althubyani, A. A., Canto, S., Pham, H., Holger, D. J., & Rey, J. (2024). Antibiotic-induced neuropsychiatric toxicity: epidemiology, mechanisms and management strategies - a narrative literature review. *Drugs in context*, 13, 2024-3-3. <https://doi.org/10.7573/dic.2024-3-3>
11. Antibiotic-resistant *Streptococcus pneumoniae*. (2024, December 17). Pneumococcal Disease. <https://www.cdc.gov/pneumococcal/php/drug-resistance/index.html>
12. Arao, Y., & Korach, K. S. (2021). The physiological role of estrogen receptor functional domains. *Essays in biochemistry*, 65(6), 867–875. <https://doi.org/10.1042/EBC20200167>
13. Arashiro, A., Andaya, R. J., Rivada, D., & McPherson, M. (2021). WPSAR celebrates World Field Epidemiology Day. *Western Pacific Surveillance and Response Journal*, 12(3), 1–4. <https://doi.org/10.5365/wpsar.2021.12.3.905>
14. Austin, P. C., White, I. R., Lee, D. S., & van Buuren, S. (2021). Missing Data in Clinical Research: A Tutorial on Multiple Imputation. *The Canadian journal of cardiology*, 37(9), 1322–1331. <https://doi.org/10.1016/j.cjca.2020.11.010>
15. B.R. Brooks, C.L. Brooks III, A.D. MacKerell, Jr., L. Nilsson, R.J. Petrella, B. Roux, Y. Won, G. Archontis, C. Bartels, S. Boresch, A. Caflisch, L. Caves, Q. Cui, A.R. Dinner, M. Feig, S. Fischer, J. Gao, M. Hodoscek, W. Im, K. Kuczera, T. Lazaridis, J. Ma, V. Ovchinnikov, E. Paci, R.W. Pastor, C.B. Post, J.Z. Pu, M. Schaefer, B. Tidor, R. M. Venable, H. L. Woodcock, X. Wu, W. Yang, D.M. York, and M. Karplus (2009) CHARMM: The Biomolecular Simulation Program. *J. Comput. Chem.* 30:1545-1614
16. Badiee, S. A., Isu, U. H., Khodadadi, E., & Moradi, M. (2023). The Alternating Access Mechanism in Mammalian Multidrug Resistance Transporters and Their Bacterial Homologs. *Membranes*, 13(6), 568. <https://doi.org/10.3390/membranes13060568>
17. Bakchi, B., Krishna, A. D., Sreecharan, E., Ganesh, V. B. J., Niharika, M., Maharshi, S., Puttagunta, S. B., Sigalapalli, D. K., Bhandare, R. R., & Shaik, A. B. (2022). An overview on applications of SwissADME web tool in the design and development of anticancer, antitubercular and antimicrobial agents: A medicinal chemist's perspective. *Journal of Molecular Structure*, 1259, 132712. <https://doi.org/10.1016/j.molstruc.2022.132712>
18. Banerjee P., Eckert O.A., Schrey A.K., Preissner R.: ProTox-II: a webserver for the prediction of toxicity of chemicals. *Nucleic Acids Res (Web server issue 2018)*
19. Banerjee P., Kemmler E., Dunkel M., Preissner R.: ProTox 3.0: a webserver for the prediction of toxicity of chemicals *Nucleic Acids Res (Web server issue 2024)*
20. Bateman, A., Martin, M., Orchard, S., Magrane, M., Agivetova, R., Ahmad, S., Alpi, E., Bowler-Barnett, E. H., Britto, R., Bursteinas, B., Bye-A-Jee, H., Coetzee, R., Cukura, A., Da Silva, A., Denny, P., Dogan, T., Ebenezer, T., Fan, J., Castro, L. G., ... Masson, P. (2020).

- UniProt: The universal protein knowledgebase in 2021. *Nucleic Acids Research*, 49(D1), D480–D489. <https://doi.org/10.1093/nar/gkaa1100>
21. Bender, Brian J., et al. “A Practical Guide to Large-scale Docking.” *Nature Protocols*, vol. 16, no. 10, Sept. 2021, pp. 4799–832. <https://doi.org/10.1038/s41596-021-00597-z>.
 22. Bento, T. P. F., Genebra, C. V. D. S., Maciel, N. M., Cornelio, G. P., Simeão, S. F. A. P., & Vitta, A. (2020). Low back pain and some associated factors: is there any difference between genders?. *Brazilian journal of physical therapy*, 24(1), 79–87. <https://doi.org/10.1016/j.bjpt.2019.01.012>
 23. Bienstock, R. J. (2015). Computational methods for fragmentbased ligand design: Growing and linking. In J. R. Morrison (Ed.), *Methods in Molecular Biology*, 1289, 119–135. https://doi.org/10.1007/9781493924868_10
 24. Bissaro, et al. (2020). The rise of molecular simulations in fragment-based drug design (FBDD): an overview. <https://pubmed.ncbi.nlm.nih.gov/32592867/>
 25. Boga, et al. (2025). Fragment-based drug discovery: A unified workflow for efficient lead generation. <https://www.ddw-online.com/fragment-based-drug-discovery-a-unified-workflow-for-efficient-lead-generation-38290-202510/>
 26. Bon, et al. (2022). Fragment-based drug discovery—the importance of high-quality molecule libraries. <https://pmc.ncbi.nlm.nih.gov/articles/PMC9627785/>
 27. Breznik, M., Ge, Y., Bluck, J. P., Briem, H., Hahn, D. F., Christ, C. D., Mortier, J., Mobley, D. L., & Meier, K. (2023). Prioritizing Small Sets of Molecules for Synthesis through in-silico Tools: A Comparison of Common Ranking Methods. *ChemMedChem*, 18(1), e202200425. <https://doi.org/10.1002/cmdc.202200425>
 28. Buccheri, R., & Rescifina, A. (2025). High-Throughput, High-Quality: Benchmarking GNINA and AutoDock Vina for Precision Virtual Screening Workflow. *Molecules*, 30(16), 3361. <https://doi.org/10.3390/molecules30163361>
 29. Buccheri, R., & Rescifina, A. (2025). High-Throughput, High-Quality: Benchmarking GNINA and AutoDock Vina for Precision Virtual Screening Workflow. *Molecules*, 30(16), 3361. <https://doi.org/10.3390/molecules30163361>
 30. Burley, S. K. et al. (2022). RCSB Protein Data Bank: Tools for visualizing and understanding biological macromolecules in 3D. *Protein Science*, 31 (12), e4482. <https://doi.org/10.1002/pro.4482>
 31. Burley, S. K. et al. (2025). Updated resources for exploring experimentally-determined PDB structures and computed structure models at the RCSB Protein Data Bank. *Nucleic Acids Research*, 53 (D1), D564–D574. <https://doi.org/10.1093/nar/gkae1091>

32. Cai, Y. et al. (2021). A comprehensive review of lignocellulosic biomass pretreatment technologies toward enzymatic hydrolysis. *Processes*, 9 (1), 71. <https://doi.org/10.3390/pr9010071>
33. Cavasotto, C. N., & Scardino, V. (2022). Machine Learning Toxicity Prediction: Latest advances by Toxicity End Point. *ACS Omega*, 7(51), 47536–47546. <https://doi.org/10.1021/acsomega.2c05693>
34. Cavasotto, C. N., & Scardino, V. (2022b). Machine Learning Toxicity Prediction: Latest advances by Toxicity End Point. *ACS Omega*, 7(51), 47536–47546. <https://doi.org/10.1021/acsomega.2c05693>
35. Centers for Disease Control and Prevention. (2024). Antibiotic-resistant *Streptococcus pneumoniae* (pneumococcal) drug resistance. <https://www.cdc.gov/pneumococcal/php/drug-resistance/index.html>
36. Chandraghatgi, et al. (2023). Streamlining Computational Fragment-Based Drug Discovery through Evolutionary Optimization Informed by Ligand-Based Virtual Prescreening. <https://www.biorxiv.org/content/10.1101/2023.11.27.568919v2.full>
37. Chang et al. (2022) A Guide to In Silico Drug Design <https://pmc.ncbi.nlm.nih.gov/articles/PMC9867171/#B3-pharmaceutics-15-00049et>
38. Chen, C., Lee, D., Yu, J., Lin, Y., & Lin, T. (2022). Recent advances in LC-MS-based metabolomics for clinical biomarker discovery. *Mass Spectrometry Reviews*. <https://doi.org/10.1002/mas.21785>
39. Chu, C. S., Simpson, J. D., O'Neill, P. M., & Berry, N. G. (2021). Machine learning – Predicting Ames mutagenicity of small molecules. *Journal of Molecular Graphics and Modelling*, 109, 108011. <https://doi.org/10.1016/j.jmgm.2021.108011>
40. Clark, F., Robb, G., Cole, D. J., & Michel, J. (2023). Comparison of Receptor-Ligand Restraint Schemes for Alchemical Absolute Binding Free Energy Calculations. *Journal of chemical theory and computation*, 19(12), 3686–3704. <https://doi.org/10.1021/acs.jctc.3c00139>
41. Cournia, et al. (2021). Free Energy Methods in Drug Discovery—Introduction. <https://pubs.acs.org/doi/10.1021/bk-2021-1397.ch001>
42. Daood, N. J., Russo, D. P., Chung, E., Qin, X., & Zhu, H. (2024). Predicting Chemical Immunotoxicity through Data-Driven QSAR Modeling of Aryl Hydrocarbon Receptor Agonism and Related Toxicity Mechanisms. *Environment & Health*, 2(7), 474–485. <https://doi.org/10.1021/envhealth.4c00026>

43. Davidson R. J. (2021). Childhood Adversity and the Brain: Harnessing the Power of Neuroplasticity. *Biological psychiatry*, 90(3), 143–144. <https://doi.org/10.1016/j.biopsych.2021.05.006>
44. De Souza Neto, L. R., Moreira-Filho, J. T., Neves, B. J., Maidana, R. L. B. R., Guimarães, A. C. R., Furnham, N., Andrade, C. H., & Silva, F. P. (2020). In silico Strategies to Support Fragment-to-Lead Optimization in Drug Discovery. *Frontiers in Chemistry*, 8, 93. <https://doi.org/10.3389/fchem.2020.00093>
45. Demczuk, W., Martin, I., Griffith, A., Lefebvre, B., McGeer, A., Tyrrell, G. J., Zhanel, G. G., Kus, J. V., Hoang, L., Minion, J., Van Caesele, P., Gad, R. R., Haldane, D., Zahariadis, G., Mead, K., Steven, L., Strudwick, L., Mulvey, M. R., ... Mulvey, M. R. (2021). Linear regression equations to predict β -lactam, macrolide, lincosamide, and fluoroquinolone MICs from molecular antimicrobial resistance determinants in *Streptococcus pneumoniae*. *Antimicrobial Agents and Chemotherapy*, 66(1), e01370-21. <https://doi.org/10.1128/aac.01370-21>
46. Dobrek L. (2023). A Synopsis of Current Theories on Drug-Induced Nephrotoxicity. *Life* (Basel, Switzerland), 13(2), 325. <https://doi.org/10.3390/life13020325>
47. Dodda, L. S., Cabeza de Vaca, I., Tirado-Rives, J., & Jorgensen, W. L. (2017). LigParGen web server: An automatic OPLS-AA parameter generator for organic ligands. *Nucleic Acids Research*, 45(W1), W331–W336. <https://doi.org/10.1093/nar/gkx312>
48. Dodda, L. S., Vilseck, J. Z., Tirado-Rives, J., & Jorgensen, W. L. (2017). 1.14*CM1A-LBCC: Localized bond-charge corrected CM1A charges for condensed-phase simulations. *The Journal of Physical Chemistry B*, 121(15), 3864–3870. <https://doi.org/10.1021/acs.jpcb.7b00272>
49. Dong, L., Qu, X., Zhao, Y., & Wang, B. (2021). Prediction of binding free energy of proteinligand complexes with a hybrid molecular mechanics/generalized Born surface area and machine learning method. *ACS Omega*, 6(48), 32938–32947. (Include in reference list with DOI from PubMed/other source)
50. Dr. Hannah M. Baumann, Dr. Joshua T. Horton, Dr. Michael M. Henry, et al. Large-scale collaborative assessment of binding free energy calculations for drug discovery using OpenFE. *ChemRxiv*. 17 March 2026.DOI: <https://doi.org/10.26434/chemrxiv-2025-7sthd/v2>
51. Dr. Irfan Alibay, Dr. Aniket Magarkar, Dr. Daniel Seeliger, et al. Evaluating the use of absolute binding free energy in the fragment optimization process.. *ChemRxiv*. 01 February 2022.DOI: <https://doi.org/10.26434/chemrxiv-2022-cw2kq-v3>
52. Duguma, T., & Dinka, H. (2024). In silico identification and characterization of drug targets in *Streptococcus pneumoniae* ATCC 700669 (serotype 23F) by subtractive genomics. *BioMed Research International*, 2024, 1–9. <https://doi.org/10.1155/2024/5917667>

53. Dunn, I., Pirhadi, S., Wang, Y., Ravindran, S., Concepcion, C., & Koes, D. R. (2024). CACHE Challenge #1: Docking with GNINA Is All You Need. *Journal of Chemical Information and Modeling*, 64(24), 9388–9396. <https://doi.org/10.1021/acs.jcim.4c01429>
54. Eberhardt, J., Santos-Martins, D., Tillack, A.F., Forli, S. (2021). AutoDock Vina 1.2.0: New Docking Methods, Expanded Force Field, and Python Bindings. *Journal of Chemical Information and Modeling*.
55. Ebrahim, R. M. A. et al. (2023). Synthesis, characterization, molecular docking, and antimicrobial activities of dinuclear nickel(II), palladium(II), and platinum(IV) complexes. *RSC Advances*, 13 , 27501–27511. <https://doi.org/10.1039/D3RA04768G>
56. Ece, A. (2023) Computer-aided drug design <https://link.springer.com/article/10.1186/s13065-023-00939-w>
57. European Bioinformatics Institute. Understanding local confidence — Evaluating AlphaFold’s predicted structures using confidence scores. 108) <https://www.ebi.ac.uk/training/online/courses/alphafold/inputs-and-outputs/evaluating-alphafolds-predicted-structures-using-confidence-scores/plddt-understanding-local-confidence/>
58. Evans, Richard, et al. “Protein Complex Prediction With AlphaFold-Multimer.” *bioRxiv* (Cold Spring Harbor Laboratory), Oct. 2021, <https://doi.org/10.1101/2021.10.04.463034>.
59. Exertier, C., et al. (2024). Fragment merging, growing, and linking identify new trypanothione reductase inhibitors for leishmaniasis. <https://pmc.ncbi.nlm.nih.gov/articles/PMC10788915/>
60. Ferla, M.P., Sánchez-García, R., Skyner, R.E. et al. Fragemstein: predicting protein–ligand structures of compounds derived from known crystallographic fragment hits using a strict conserved-binding-based methodology. *J Cheminform* 17, 4 (2025). <https://doi.org/10.1186/s13321-025-00946-0>
61. Fierro, Fabrizio, et al. “Inhibiting a Promiscuous GPCR: Iterative Discovery of Bitter Taste Receptor Ligands.” *Cellular and Molecular Life Sciences*, vol. 80, no. 4, Apr. 2023, p. 114. <https://doi.org/10.1007/s00018-023-04765-0>.
62. Fisher, J. E., Guha, A., Heller, W., & Miller, G. A. (2020). Extreme-groups designs in studies of dimensional phenomena: Advantages, caveats, and recommendations. *Journal of abnormal psychology*, 129(1), 14–20. <https://doi.org/10.1037/abn0000480>
63. Forli, S., & Botta, M. (2007). Lennard-Jones potential and dummy atom settings to overcome the AUTODOCK limitation in treating flexible ring systems. *Journal of chemical information and modeling*, 47(4), 1481-1492.
64. Forli, S., & Olson, A. J. (2012). A force field with discrete displaceable waters and desolvation entropy for hydrated ligand docking. *Journal of medicinal chemistry*, 55(2), 623-638.

65. Forli, S., Huey, R., Pique, M. E., Sanner, M. F., Goodsell, D. S., & Olson, A. J. (2016). Computational protein–ligand docking and virtual drug screening with the AutoDock suite. *Nature protocols*, 11(5), 905-919.
66. Forouzesh, F., et al. (2021). MM/GBSA application for absolute binding free energy calculation in protein–ligand systems. (Include proper bibliographic details and DOI from your own source)
67. *Frontiers in Pharmacology*. (n.d.). *Frontiers Media S.A.*
<https://www.frontiersin.org/journals/pharmacology/articles>
68. *Frontiers in Public Health*. (n.d.). *Frontiers Media S.A.*
<https://www.frontiersin.org/journals/public-health/articles>
69. Gamil, A., Lalas, M. Y., Capeding, M. R. Z., Ong-Lim, A. L. T., Bunyi, M. A. C., & Claveria, A. M. (2021). A narrative review of pneumococcal disease in children in the Philippines. *Infectious Diseases and Therapy*, 10(2), 699–718. <https://doi.org/10.1007/s40121-021-00434-6>
70. GBD 2021 Lower Respiratory Infections and Antimicrobial Resistance Collaborators et al. (2024). Global, regional, and national incidence and mortality burden of non-COVID-19 lower respiratory infections and aetiologies, 1990–2021: A systematic analysis from the Global Burden of Disease Study 2021. *The Lancet Infectious Diseases*, 24 (9), 974–1002.
[https://doi.org/10.1016/S1473-3099\(24\)00176-2](https://doi.org/10.1016/S1473-3099(24)00176-2)
71. Genheden, S., & Ryde, U. (2021). The MM/PBSA and MM/GBSA methods to estimate ligandbinding affinities. *Expert Opinion on Drug Discovery*, 16(5), 449–461.
<https://doi.org/10.1080/17460441.2020.1860033> (If needed for reference list)
72. Ghane Shahrbaf, F., & Assadi, F. (2015). Drug-induced renal disorders. *Journal of renal injury prevention*, 4(3), 57–60. <https://doi.org/10.12861/jrip.2015.12>
73. Gibertoni, D., Adja, K. Y. C., Golinelli, D., Reno, C., Regazzi, L., Lenzi, J., Sanmarchi, F., & Fantini, M. P. (2021). Patterns of COVID-19 related excess mortality in the municipalities of Northern Italy during the first wave of the pandemic. *Health & place*, 67, 102508.
<https://doi.org/10.1016/j.healthplace.2021.102508>
74. Goodsell, D. S. et al. (2022). RCSB Protein Data Bank resources for structure-facilitated discovery and design. *Structure*, 30 (10), 1235–1246. <https://doi.org/10.1016/j.str.2021.10.003>
75. Guo, W., Liu, J., Dong, F., Song, M., Li, Z., Khan, M. K. H., Patterson, T. A., & Hong, H. (2023). Review of machine learning and deep learning models for toxicity prediction. *Experimental Biology and Medicine*, 248(21), 1952–1973. <https://doi.org/10.1177/15353702231209421>
76. H.M. Berman, J. Westbrook, Z. Feng, G. Gilliland, T.N. Bhat, H. Weissig, I.N. Shindyalov, P.E. Bourne, The Protein Data Bank (2000) *Nucleic Acids Research* 28: 235-242
<https://doi.org/10.1093/nar/28.1.235>.

77. H.M. Berman, K. Henrick, H. Nakamura Announcing the worldwide Protein Data Bank (2003) *Nature Structural Biology* 10:980 <https://doi.org/10.1038/nsb1203-980>.
78. Han, H., Shaker, B., Lee, J. H., Choi, S., Yoon, S., Singh, M., Basith, S., Cui, M., Ahn, S., An, J., Kang, S., Yeom, M. S., & Choi, S. (2025). Employing automated machine learning (AutoML) methods to facilitate the in silico ADMET properties prediction. *Journal of Chemical Information and Modeling*, 65(7), 3215–3225. <https://doi.org/10.1021/acs.jcim.4c02122>
79. Han, J. et al. (2024). Rapid discovery of antimicrobial and antimalarial agents from natural product fragments. *Separations*, 11 (7), 194. <https://doi.org/10.3390/separations11070194>
80. Hasan, M. M., Rahman, M. M., & Hossain, M. M. (2025). Molecular docking and dynamics in protein serine/threonine kinase drug discovery. *Frontiers in Pharmacology*, 16, 1696204. <https://www.frontiersin.org/articles/10.3389/fphar.2025.1696204/full>
81. Holcomb, Matthew, et al. “Evaluation of AlphaFold2 Structures as Docking Targets.” *Protein Science*, vol. 32, no. 1, Dec. 2022, p. e4530. <https://doi.org/10.1002/pro.4530>.
82. Huang, S.-Y., & Zou, X. (2010). Advances and challenges in protein-ligand docking . *International Journal of Molecular Sciences*, 11 (8), 3016–3034. <https://doi.org/10.3390/ijms11083016>
83. Hukerikar, N., Hingorani, A. D., Asselbergs, F. W., Finan, C., & Schmidt, A. F. (2024). Prioritising genetic findings for drug target identification and validation. *Atherosclerosis*, 390, 117462. <https://doi.org/10.1016/j.atherosclerosis.2024.117462>
84. Ibezim, A. et al. (2022). Fragment-based virtual screening discovers potential new Plasmodium PI4KIII β ligands. *BMC Chemistry*, 16 , 19. <https://doi.org/10.1186/s13065-022-00812-2>
85. Imieje, V. O., & Ajayi-Ojo, E. O. (2025). In silico ADMET profiling and post-docking analysis of phytoconstituents... *Journal of Applied Science and Environmental Management*, 29(9), 2768–2781. <https://doi.org/10.4314/jasem.v29i9.10>
86. J. Lee, X. Cheng, J.M. Swails, M.S. Yeom, P.K. Eastman, J.A. Lemkul, S. Wei, J. Buckner, J.C. Jeong, Y. Qi, S. Jo, V.S. Pande, D.A. Case, C.L. Brooks III, A.D. MacKerell Jr, J.B. Klauda, and W. Im (2016) CHARMM-GUI Input Generator for NAMD, GROMACS, AMBER, OpenMM, and CHARMM/OpenMM Simulations using the CHARMM36 Additive Force Field. *J. Chem. Theory Comput.* 12:405-413
87. James C. Phillips, David J. Hardy, Julio D. C. Maia, John E. Stone, Joao V. Ribeiro, Rafael C. Bernardi, Ronak Buch, Giacomo Fiorin, Jerome Henin, Wei Jiang, Ryan McGreevy, Marcelo C. R. Melo, Brian K. Radak, Robert D. Skeel, Abhishek Singharoy, Yi Wang, Benoît Roux, Aleksei Aksimentiev, Zaida Luthey-Schulten, Laxmikant V. Kale, Klaus Schulten, Christophe Chipot, and

- Emad Tajkhorshid. Scalable molecular dynamics on CPU and GPU architectures with NAMD. *Journal of Chemical Physics*, 153:044130, 2020. doi:10.1063/5.0014475
88. Jiang, D., Du, H., Zhao, H., Deng, Y., Wu, Z., Wang, X., ... & Hsieh, C. (2024). Assessing the performance of MM/PBSA and MM/GBSA methods: Reliability of binding affinities and poses for RNA–ligand complexes. *Physical Chemistry Chemical Physics*, 26, 10323–10335. <https://doi.org/10.1039/D3CP04366E>
89. Jin, P., Duan, X., Huang, Z., Dong, Y., Zhu, J., Guo, H., Tian, H., Zou, C. G., & Xie, K. (2025). Nuclear receptors in health and disease: signaling pathways, biological functions and pharmaceutical interventions. *Signal transduction and targeted therapy*, 10(1), 228. <https://doi.org/10.1038/s41392-025-02270-3>
90. Jin, W., et al. (2020). Next-generation molecular analysis and drug screening. PubMed Central. <https://pmc.ncbi.nlm.nih.gov/articles/PMC7778908/>
91. Jorgensen, W. L., & Tirado-Rives, J. (2005). Potential energy functions for atomic-level simulations of water and organic and biomolecular systems. *Proceedings of the National Academy of Sciences of the United States of America*, 102(19), 6665–6670. <https://doi.org/10.1073/pnas.0408037102>
92. Jumper, J. et al. “Highly accurate protein structure prediction with AlphaFold.” *Nature*, 596, pages 583–589 (2021). DOI: 10.1038/s41586-021-03819-2.
93. Kagami, L., Wilter, A., Diaz, A., & Vranken, W. (2023). The ACPYPE web server for small-molecule MD topology generation. *Bioinformatics (Oxford, England)*, 39(6), btad350. <https://doi.org/10.1093/bioinformatics/btad350>
94. Kalas, M. A., Chavez, L., Leon, M., Taweasedt, P. T., & Surani, S. (2021). Abnormal liver enzymes: A review for clinicians. *World journal of hepatology*, 13(11), 1688–1698. <https://doi.org/10.4254/wjh.v13.i11.1688>
95. Kalogeropoulos, K. et al. (2024). A comparative study of protein structure prediction tools for challenging targets: Snake venom toxins. *Toxicon*, 238, 107559. <https://pubmed.ncbi.nlm.nih.gov/38113945/>
96. Karelina, Masha, et al. “How Accurately Can One Predict Drug Binding Modes Using AlphaFold Models?” *eLife*, vol. 12, Dec. 2023, <https://doi.org/10.7554/elife.89386.2>.
97. Katzberger, P., & Riniker, S. (2023). Implicit solvent approach based on generalized Born and transferable graph neural networks for molecular dynamics simulations. *The Journal of Chemical Physics*, 158(20). <https://doi.org/10.1063/5.0147027>

98. Kaur, R., Fuji, N., & Pichichero, M. E. (2021). Dynamic changes in otopathogens after PCV13 vaccination. *European Journal of Clinical Microbiology & Infectious Diseases*, 41(1), 37–44. <https://doi.org/10.1007/s10096-021-04324-0>
99. Kerr, M., Ackland, G. J., Marenduzzo, D., Brandani, G. B., & Pruteanu, C. G. (2024). Can the AMOEBA forcefield be used for high pressure simulations? The extreme case of methane and water. *The Journal of Chemical Physics*, 161(5). <https://doi.org/10.1063/5.0211222>
100. Kim, et al. (2020). CHARMM-GUI Free Energy Calculator for Absolute and Relative Ligand Solvation and Binding Free Energy Simulations. <https://pmc.ncbi.nlm.nih.gov/articles/PMC7658063/>
101. Kim, S. et al. (2021). Exploring chemical information in PubChem. *Current Protocols*, 2(3), e217. <https://doi.org/10.1002/cpz1.217>
102. Kiss, R., Sandor, M., & Szalai, F. A. (2012). <http://Mcule.com>: a public web service for drug discovery. *Journal of Cheminformatics*, 4(S1). <https://doi.org/10.1186/1758-2946-4-s1-p17>
103. Klimoszek, D., Jeleń, M., Dołowy, M., & Morak-Młodawska, B. (2024). Study of the Lipophilicity and ADMET Parameters of New Anticancer Diquinotiazines with Pharmacophore Substituents. *Pharmaceuticals*, 17(6), 725. <https://doi.org/10.3390/ph17060725>
104. Konaklieva, M. I., & Plotkin, B. J. (2023). Fragment-based lead discovery strategies in antimicrobial drug discovery. *Antibiotics*, 12(2), 315. <https://doi.org/10.3390/antibiotics12020315>
105. Kores, K., Kolenc, Z., Furlan, V., & Bren, U. (2022). Inverse molecular docking elucidating the anticarcinogenic potential of the hop natural product xanthohumol and its metabolites. *Foods*, 11(9), 1253. <https://www.mdpi.com/2304-8158/11/9/1253>
106. Kralj, S., Jukič, M., & Bren, U. (2023). Molecular Filters in Medicinal Chemistry. *Encyclopedia*, 3(2), 501–511. <https://doi.org/10.3390/encyclopedia3020035>
107. Krishna. (2025). Computational drug likeness profiling and chemical space analysis... <https://doi.org/10.26434/chemrxiv-2025-qdp8w>
108. Künili, İ. E., Akdeniz, V., Akpınar, A., Budak, Ş. Ö., Curiel, J. A., Guzel, M., Karagözlü, C., Kasikci, M. B., Caruana, G. P. M., Starowicz, M., Humblot, C., Keyvan, E., Chassard, C., Pracer, S., Vergères, G., & Kesenkaş, H. (2025). Bioactive compounds in fermented foods: a systematic narrative review. *Frontiers in Nutrition*, 12, 1625816. <https://doi.org/10.3389/fnut.2025.1625816>
109. Kyle Swanson, Parker Walther, Jeremy Leitz, Souhrid Mukherjee, Joseph C Wu, Rabindra V Shivnaraine, James Zou, ADMET-AI: a machine learning ADMET platform for evaluation of large-scale chemical libraries, *Bioinformatics*, Volume 40, Issue 7, July 2024, btae416, <https://doi.org/10.1093/bioinformatics/btae416>

110. Landrum, G. (2025, August 15). BRICS tutorial. RDKit. <https://greglandrum.github.io/rdkit-blog/posts/2025-08-15-BRICS-tutorial.html#:~:text=On%20this%20page,full/10.1002/cmdc.200800178>
111. Landrum, G., Tosco, P., Kelley, B., Rodriguez, R., Cosgrove, D., Vianello, R., Sriniker, Gedeck, P., Jones, G., Kawashima, E., NadineSchneider, Nealschneider, D., Dalke, A., Tadhurst-Cdd, Swain, M., Cole, B., Turk, S., Savelev, A., Maeder, N., . . . Lehtivarjo, J. (2025). rdkit/rdkit: 2025_09_3 (Q3 2025) Release. Zenodo. <https://doi.org/10.5281/zenodo.17746401>
112. Lenselink, E. B., & Stouten, P. F. W. (2021). Multitask machine learning models for predicting lipophilicity (logP) in the SAMPL7 challenge. *Journal of Computer-Aided Molecular Design*, 35(8), 901–909. <https://doi.org/10.1007/s10822-021-00405-6>
113. León, R., SotoDelgado, J., Montero, E., & Vargas, M. (2021). Development of computational approaches with a fragmentbased drug design strategy: In silico Hsp90 inhibitors discovery. *International Journal of Molecular Sciences*, 22(24), 13226. <https://doi.org/10.3390/ijms222413226>
114. Lesk. (2025). Bioinformatics. Britannica. <https://www.britannica.com/science/bioinformatics>
115. Li, J., Cheng, G., Qin, X., & Liu, J. (2025). Streptococcus pneumoniae β -lactam resistance: Trends and molecular drivers. *Clinical Microbiology Reviews*, 38(3), e00082-25. <https://doi.org/10.1128/cmr.00082-25>
116. Li, L., Ma, J., Yu, Z., Li, M., Zhang, W., & Sun, H. (2022). Epidemiological characteristics and resistance mechanisms of *S. pneumoniae*. *Microbiological Research*, 266, 127221. <https://doi.org/10.1016/j.micres.2022.127221>
117. Liao, et al. (2022). In Silico Methods for Identification of Potential Active Sites of Therapeutic Targets. <https://www.mdpi.com/1420-3049/27/20/7103>
118. Lui, R., Guan, D., & Matthews, S. (2020). A comparison of molecular representations for lipophilicity quantitative structure–property relationships with results from the SAMPL6 logP Prediction Challenge. *Journal of Computer-Aided Molecular Design*, 34(5), 523–534. <https://doi.org/10.1007/s10822-020-00279-0>
119. Lukashina, N., Alenicheva, A., Vlasova, E., Kondiukov, A., Khakimova, A., Magerramov, E., Churikov, N., & Shpilman, A. (2020, November 24). Lipophilicity Prediction with Multitask Learning and Molecular Substructures Representation. arXiv.org. <https://arxiv.org/abs/2011.12117>
120. Lundquist, K. P., Panchal, V., Gotfredsen, C. H., Brenk, R., & Clausen, M. H. (2021). Fragment-Based drug discovery for RNA targets. *ChemMedChem*, 16(17), 2588–2603. <https://doi.org/10.1002/cmdc.202100324>

121. Luo et al. (2023) Benchmarking Reverse Docking through AlphaFold2 Human Proteome <https://doi.org/10.1101/2023.12.16.572027>
122. Luo, Q. et al. (2025). AutoRevDock: An open-source toolkit for scalable reverse docking in drug discovery. Protein Science. <https://pubmed.ncbi.nlm.nih.gov/41123282/>
123. Lv Y, Luo Y-Y, Ren H-W, Li C-J, Xiang Z-X and Luan Z-L (2022) The role of pregnane X receptor (PXR) in substance metabolism. Front. Endocrinol. 13:959902. doi: <https://doi.org/10.3389/fendo.2022.959902>
124. Manyana et al. (2021). Drug Resistance Genotyping <https://doi.org/10.3390/v13061125>
125. Marbán-González, A., Ramírez-Cid, V., Cristóbal-Ramírez, A., & Medina-Franco, J. L. (2025). Exploiting PubChem and other public databases for virtual screening in 2025: What are the latest trends? Expert Opinion on Drug Discovery . <https://doi.org/10.1080/17460441.2025.2558161>
126. Martis, et al. (2025). Ten quick tips to perform meaningful and reproducible molecular docking calculations, <https://pmc.ncbi.nlm.nih.gov/articles/PMC12064015/>
127. Mary E Gilbert, Katherine L O'Shaughnessy, Marta Axelstad, Regulation of Thyroid-disrupting Chemicals to Protect the Developing Brain, Endocrinology, Volume 161, Issue 10, October 2020, bqaa106, <https://doi.org/10.1210/endocr/bqaa106>
128. McCarthy, M. et al. (2022) Reverse molecular docking and deep-learning to make predictions of receptor activity for neurotoxicology <https://www.sciencedirect.com/science/article/pii/S2468111322000263>
129. Medina-Franco, et al. (2022) . Chemical Multiverse: An Expanded View of Chemical Space. <https://pmc.ncbi.nlm.nih.gov/articles/PMC9787733/>
130. Mohanty, S. et al. (2023). Trends in Streptococcus pneumoniae antimicrobial resistance in US children: A multicenter evaluation. Open Forum Infectious Diseases, 10 (3), ofad098. <https://doi.org/10.1093/ofid/ofad098>
131. Mohanty, S., Feemster, K., Yu, K. C., Watts, J. A., & Gupta, V. (2023). Trends in Streptococcus pneumoniae antimicrobial resistance in U.S. children. Open Forum Infectious Diseases, 10(3), ofad098. <https://doi.org/10.1093/ofid/ofad098>
132. Molecular docking for Computer-Aided drug design. (2021). In Elsevier eBooks. <https://doi.org/10.1016/c2019-0-04960-6>
133. Moreno, et al. (2020). Molecular characterization of Latin American invasive Streptococcus pneumoniae serotype 19A isolates. <https://www.sciencedirect.com/science/article/abs/pii/S0264410X20303856>

134. Mr. Yathu Krishna Y K. Computational Drug Likeness Profiling and Chemical Space Analysis of Bioactive Microbial Secondary Metabolites via RDKit, PCA, and Tanimoto-Based Fingerprint Clustering . ChemRxiv. 06 August 2025.DOI: <https://doi.org/10.26434/chemrxiv-2025-qdp8w>
135. N, N. K., & K, N. K. (2024). Target identification and validation in research. World Journal of Biology Pharmacy and Health Sciences, 17(3), 107–117. <https://doi.org/10.30574/wjbphs.2024.17.3.0116>
136. Narang. (2023). Computational Biology: The Intersection of Computer Science and Life Sciences.<https://www.longdom.org/articles/computational-biology-the-intersection-of-computer-science-and-life-sciences-100196.html>
137. Narciso, A. R., Dookie, R., Nannapaneni, P., Normark, S., & Henriques-Normark, B. (2024). Streptococcus pneumoniae epidemiology, pathogenesis and control. Nature Reviews Microbiology, 23. <https://doi.org/10.1038/s41579-024-01116-z>
138. Neto, et al. (2020). In silico Strategies to Support Fragment-to-Lead Optimization in Drug Discovery. <https://www.frontiersin.org/journals/chemistry/articles/10.3389/fchem.2020.00093/full>
139. Niu, Z., Xiao, X., Wu, W. et al. PharmaBench: Enhancing ADMET benchmarks with large language models. Sci Data 11, 985 (2024). <https://doi.org/10.1038/s41597-024-03793-0>
140. Ntim, et al. (2025). Relationship between pneumococcal serotypes and antimicrobial resistance: A systematic review and meta-analysis. <https://www.sciencedirect.com/science/article/pii/S2213716525001663>
141. Ochoa, et al. (2024). Epidemiological characteristics and antibiotic resistance mechanisms of Streptococcus pneumoniae: An updated review. https://www.sciencedirect.com/science/article/pii/S0944501322002610?utm_so
142. Ogunjobi, T. T., Okorie, I. C., Gigam-Ozuzu, C. D., Olorunleke, J. V., Ogunleye, F. I., Irimoren, E. O., Atanda, D. O., Okafor, A. M.-A., Agbo, C. E., Okunbi, F. O., Umoren, O. D., Adidu, A. D., & Ojo, E. O. (2025). Bioinformatics tools in protein analysis: Structure prediction, interaction modelling, and function relationship. European Journal of Sustainable Development Research, 9(3), em0298. <https://www.ejosdr.com/article/bioinformatics-tools-in-protein-analysis-structure-prediction-interaction-modelling-and-function-16340>
143. Olson et al. (2024). Introducing the Bacterial and Viral Bioinformatics Resource Center (BV-BRC) resource. Journal of Comparative Virology and Informatics. <https://www.jcvi.org/publications/introducing-bacterial-and-viral-bioinformatics-resource-center-bv-brc-resource>

144. Olson, R. D., Assaf, R., Brettin, T., Conrad, N., Cucinell, C., Davis, J. J., ... Stevens, R. L. (2022). Introducing the BV-BRC. *Nucleic Acids Research*, 51(D1), D678–D689. <https://doi.org/10.1093/nar/gkac1003>
145. PacaYang. (2025, July 11). rdkit.Chem.rdDetermineBonds issue in receptor preparation [Online forum post]. GitHub. <https://github.com/forlilab/Meeko/issues/371>
146. Pal, S., Chaudhary, M., & Kala, K. (2025). Autodock & Autodock Vina: Development, Capabilities, & applications in Molecular Docking. In *International Journal of Pharmaceutical Sciences*, International Journal of Pharmaceutical Sciences (Vol. 3, Issue 5, pp. 3005–3019) [Journal-article]. <https://doi.org/10.5281/zenodo.15458275>
147. Pan American Health Organization. (n.d.). Pneumococcus . <https://www.paho.org/en/topics/pneumococcus>
148. Patel, R. S., & Singh, M. (2022). Analysis of microbial bioinformatics and interactions. *Nucleic Acids Research*, 50(4), 1202–1214. <https://doi.org/10.1093/nar/gkaf1254>
149. PatSnap Synapse. (n.d.). What is computer-aided drug design (CADD) and what is its current scenario in drug discovery? <https://synapse.patsnap.com/article/what-is-computer-aided-drug-design-cadd-what-is-its-current-scenario-in-drug-discovery>
150. Perazella, M. A., & Rosner, M. H. (2022). Drug-Induced Acute Kidney Injury. *Clinical journal of the American Society of Nephrology : CJASN*, 17(8), 1220–1233. <https://doi.org/10.2215/CJN.11290821>
151. Peter Eastman, Raimondas Galvelis, Raúl P. Peláez, Charles R. A. Abreu, Stephen E. Farr, Emilio Gallicchio, Anton Gorenko, Michael M. Henry, Frank Hu, Jing Huang, Andreas Krämer, Julien Michel, Joshua A. Mitchell, Vijay S. Pande, João PGLM Rodrigues, Jaime Rodríguez-Guerra, Andrew C. Simmonett, Sukrit Singh, Jason Swails, Philip Turner, Yuanqing Wang, Ivy Zhang, John D. Chodera, Gianni De Fabritiis, and Thomas E. Markland *The Journal of Physical Chemistry B* **2024** /28 (1), 109-116 DOI: <https://doi.org/10.1021/acs.jpcc.3c06662>
152. Pettersen, et al. (2020). UCSF ChimeraX: Structure visualization for researchers, educators, and developers. <https://onlinelibrary.wiley.com/doi/full/10.1002/pro.3943>
153. Philippine Statistics Authority. (2025). 2024 causes of death in the Philippines (provisional as of 31 December2024). <https://psa.gov.ph/content/2024-causes-death-philippines-provisional-31-december-2024>
154. Philippine Statistics Authority. (2025). 2024 causes of death in the Philippines (provisional as of 31 July 2025) . <https://psa.gov.ph/content/2024-causes-death-philippines-provisional-31-july-2025>

155. Pneumococcus. (2024, August 30). Pan American Health Organization. <https://www.paho.org/en/topics/pneumococcus>
156. Posit team (2025). RStudio: Integrated Development Environment for R. Posit Software, PBC, Boston, MA. URL <http://www.posit.co/>
157. Priyanka Banerjee, Emanuel Kemmler, Mathias Dunkel, Robert Preissner, ProTox 3.0: a webserver for the prediction of toxicity of chemicals, *Nucleic Acids Research*, Volume 52, Issue W1, 5 July 2024, Pages W513–W520, <https://doi.org/10.1093/nar/gkae303>
158. Priyanka Banerjee, Emanuel Kemmler, Mathias Dunkel, Robert Preissner, ProTox 3.0: a webserver for the prediction of toxicity of chemicals, *Nucleic Acids Research*, Volume 52, Issue W1, 5 July 2024, Pages W513–W520, <https://doi.org/10.1093/nar/gkae303>
159. R Core Team (2021). R: A language and environment for statistical computing. R Foundation for Statistical Computing, Vienna, Austria. URL <https://www.R-project.org/>.
160. Radkowski, P., Oszytko, J., Sobolewski, K., Trachte, F., Onichimowski, D., & Majewska, M. (2025). The Effect of Antibiotics on the Nervous System: Importance for Anesthesiology and Intensive Care. *Antibiotics*, 14(6), 622. <https://doi.org/10.3390/antibiotics14060622>
161. Raveh, Barak, et al. “Bayesian Metamodeling of Complex Biological Systems Across Varying Representations.” *Proceedings of the National Academy of Sciences*, vol. 118, no. 35, Aug. 2021, <https://doi.org/10.1073/pnas.2104559118>.
162. Ravnik, V., Jukič, M., Furlan, V., Maver, U., Rožanc, J., & Bren, U. (2025). Uncovering molecular targets of phytocannabinoids. *Frontiers in Pharmacology*, 16, 1611461. <https://doi.org/10.3389/fphar.2025.1611461>
163. Reddy, V. B., Bhavani, P. S., & St John, K. (2003). Silver and copper zeolites: Bactericidal and virucidal materials. *Journal of Materials Chemistry*, 13 (10), 2581–2587. <https://doi.org/10.1039/B306515N>
164. Reiter, D. A., Adelnia, F., Cameron, D., Spencer, R. G., & Ferrucci, L. (2021). Parsimonious modeling of skeletal muscle perfusion: Connecting the stretched exponential and fractional Fickian diffusion. *Magnetic resonance in medicine*, 86(2), 1045–1057. <https://doi.org/10.1002/mrm.28766>
165. Robertson, Michael J., et al. “GemSpot: A Pipeline for Robust Modeling of Ligands Into Cryo-EM Maps.” *Structure*, vol. 28, no. 6, May 2020, pp. 707-716.e3. <https://doi.org/10.1016/j.str.2020.04.018>.
166. Russomando, G. et al. (2025). *Streptococcus pneumoniae* carriage, antimicrobial resistance, and serotype distribution in children and adults from Paraguay in the post-vaccinal era. *Frontiers in Public Health*, 13, 1584857. <https://doi.org/10.3389/fpubh.2025.1584857>

167. S. Jo, T. Kim, V.G. Iyer, and W. Im (2008) CHARMM-GUI: A Web-based Graphical User Interface for CHARMM. *J. Comput. Chem.* 29:1859-1865
168. Saller, Julia, et al. "Identification and Quantification of Kukoamine a and Kukoamine B as Novel M-opioid Receptor Agonists in Potato and Other Solanaceous Plants." *Food Chemistry*, vol. 427, no. 8, June 2023, p. 136637. <https://doi.org/10.1016/j.foodchem.2023.136637>.
169. Salo-Ahen et al. (2021). Molecular Dynamics Simulations <https://doi.org/10.3390/pr9010071>
170. Salo-Ahen, O. M. H., Alanko, I., Bhadane, R., Bonvin, A. M. J. J., Honorato, R. V., Hossain, S., Juffer, A. H., Kabedev, A., Lahtela-Kakkonen, M., Larsen, A. S., Lescrinier, E., Marimuthu, P., Mirza, M. U., Mustafa, G., Nunes-Alves, A., Pantsar, T., Saadabadi, A., Singaravelu, K., & Vanmeert, M. (2020). Molecular dynamics simulations in drug discovery and pharmaceutical development. *Processes*, 9(1), 71. <https://doi.org/10.3390/pr9010071>
171. Santiago-McRae, E., Ebrahimi, M., Sandberg, J. W., Brannigan, G., & Hénin, J. (2023). Computing Absolute binding affinities by Streamlined Alchemical Free Energy Perturbation (SAFEP) [Article v1.0]. *Living Journal of Computational Molecular Science*, 5(1), 2067. <https://doi.org/10.33011/livecoms.5.1.2067>
172. Santos, C. a. L. D., De Araújo Monteiro, A. A., Da Silva Teles, B. R., Kamdem, J., Duarte, A. E., Alqhtani, H. A., Abukhadra, M. R., Allam, A. A., & Ibrahim, M. (2025). Integrative biochemical and computational analysis of leflunomide-induced redox alterations in *Nauphoeta cinerea*. *Scientific Reports*, 16(1), 3298. <https://doi.org/10.1038/s41598-025-33177-2>
173. Santos-Martins, D., Eberhardt, J., Bianco, G., Solis-Vasquez, L., Ambrosio, F. A., Koch, A., & Forli, S. (2019). D3R Grand Challenge 4: prospective pose prediction of BACE1 ligands with AutoDock-GPU. *Journal of Computer-Aided Molecular Design*, 33(12), 1071-1081.
174. Santos-Martins, D., Forli, S., Ramos, M. J., & Olson, A. J. (2014). AutoDock4Zn: an improved AutoDock force field for small-molecule docking to zinc metalloproteins. *Journal of chemical information and modeling*, 54(8), 2371-2379.
175. Saponaro, F., Sestito, S., Runfola, M., Rapposelli, S., & Chiellini, G. (2020). Selective Thyroid Hormone Receptor-Beta (TRB) Agonists: New perspectives for the treatment of Metabolic and Neurodegenerative Disorders. *Frontiers in Medicine*, 7, 331. <https://doi.org/10.3389/fmed.2020.00331>
176. Sarah Da Won Bae, Romario Nguyen, Liang Qiao, Jacob George, Role of the constitutive androstane receptor (CAR) in human liver cancer, *Biochimica et Biophysica Acta (BBA) - Reviews on Cancer*, Volume 1875, Issue 2, 2021, 188516, ISSN 0304-419X,

<https://doi.org/10.1016/j.bbcan.2021.188516>.

(<https://www.sciencedirect.com/science/article/pii/S0304419X21000159>)

177. Sarthak Pal, Meenu Chaudhary*, Kanishak Kala, Autodock & Autodock Vina: Development, Capabilities, & Applications in Molecular Docking, *Int. J. of Pharm. Sci.*, 2025, Vol 3, Issue 5, 3005-3019. <https://doi.org/10.5281/zenodo.15458275>
178. Scopus – Document details. (n.d.). <https://www.scopus.com/pages/publications/85123651244>
179. Severoglu, Y. B., Yuksel, B., Sucu, C., Aral, N., Uversky, V. N., & Coskuner-Weber, O. (2025). Implicit Solvent Models and Their Applications in Biophysics. *Biomolecules*, 15(9), 1218. <https://doi.org/10.3390/biom15091218>
180. Shaik, M. R., Patan, A., & Yadala, P. C. (2025). A review of SwissADME in Morden Medicinal Chemistry: Insights on Anti-Cancer, Anti-Tubercular, and Anti-Microbial Agents. *International Journal of Current Trends in Pharmaceutical Research*, 13(02), 90–96. <https://doi.org/10.30904/j.ijctpr.2025.4865>
181. Shan, J., & Ji, C. (2020). MolOpt: A Web Server for Drug Design using Bioisosteric Transformation. *Current computer-aided drug design*, 16(4), 460–466. <https://doi.org/10.2174/1573409915666190704093400>
182. Shawrob, K. S. M., Dhariwal, A., Salvadori, G., Gladstone, R. A., & Junges, R. (2025). Global molecular epidemiology of antibiotic resistance determinants. *Microbial Genomics*, 11(7). <https://doi.org/10.1099/mgen.0.001444>
183. Shukla, M. et al. (2025). BV-BRC: a unified bacterial and viral bioinformatics resource. *Nucleic Acids Research*. <https://academic.oup.com/nar/advance-article/doi/10.1093/nar/gkaf1254/8326459?login=false>
184. Shukla, M., Wattam, A. R., Aleman, A., Bhattacharya, R., Bowers, N., Brettin, T., ... Stevens, R. L. (2025). BV-BRC: A unified bacterial and viral bioinformatics resource. *Nucleic Acids Research*. <https://doi.org/10.1093/nar/gkaf1254>
185. Sia, S. B., Lagrada, M. L., Gayeta, J. M., Masim, M. A. L., Abad, J. P., Magbanua, M. A., & Ablola, F. B. (2021). Serotype distribution and antimicrobial resistance in the Philippines. <https://doi.org/10.5365/wpsar-2021.12.4.834>
186. Sinha, A. (2025, October 19). The overlooked role of intrinsic water in protein–ligand binding. *LiteFold*. <https://blog.litefold.ai/intrinsicwater/#:~:text=Water%20molecules%20are%20dynamic%20participants,maintaining%20the%20overall%20protein%20architecture>

187. Soffer, A., Viswas, S. J., Alon, S., Rozenberg, N., Peled, A., Piro, D., Vilenchik, D., & Akabayov, B. (2024). MolOptimizer: A molecular optimization toolkit for fragmentbased drug design. *Molecules*, 29(1), 276. <https://doi.org/10.3390/molecules29010276>
188. Spiegel, J.O., Durrant, J.D. AutoGrow4: an open-source genetic algorithm for de novo drug design and lead optimization. *J Cheminform* 12, 25 (2020). <https://doi.org/10.1186/s13321-020-00429-4>
189. Stafford, Kate A., et al. “AtomNet PoseRanker: Enriching Ligand Pose Quality for Dynamic Proteins in Virtual High-Throughput Screens.” *Journal of Chemical Information and Modeling*, vol. 62, no. 5, Mar. 2022, pp. 1178–89. <https://doi.org/10.1021/acs.jcim.1c01250>.
190. Su Hyung Park, Youyoung Kim, Jae Sun Ra, Min woo Wie, Mi-Sun Kang, Sukhyun Kang, Kyungjae Myung, Kyoo-young Lee, Timely termination of repair DNA synthesis by ATAD5 is important in oxidative DNA damage-induced single-strand break repair, *Nucleic Acids Research*, Volume 49, Issue 20, 18 November 2021, Pages 11746–11764, <https://doi.org/10.1093/nar/gkab999>
191. Sungurlu, S., & Balk, R. A. (2024). The Role of Biomarkers in the Diagnosis and Management of Pneumonia. *Infectious Disease Clinics of North America*, 38(1), 35–49. <https://doi.org/10.1016/j.idc.2023.12.005>
192. Talevi, A. (2024) Computer-Aided Drug Discovery and Design: Recent Advances and Future Prospects <https://pubmed.ncbi.nlm.nih.gov/37676590/>
193. Tan, J., Zhang, Y., & Wang, J. (2024). RmsdXNA: RMSD prediction of nucleic acid–ligand docking poses using machine learning. *Briefings in Bioinformatics*, 25(3), bbae166. <https://doi.org/10.1093/bib/bbae166>
194. Tang, et al. (2024). Recent Advances in Automated Structure-Based De Novo Drug Design. <https://pmc.ncbi.nlm.nih.gov/articles/PMC10966644/>
195. Taylor, M., & Ho, J. (2023). MM/GBSA prediction of relative binding affinities of carbonic anhydrase inhibitors: Effect of atomic charges and comparison with AutoDock4Zn. *Journal of ComputerAided Molecular Design*, 37(4), 167–182. <https://doi.org/10.1007/s10822-023-00499-0>
196. Tian S. et al. (2022). Study on subacute inhalation toxicity and offspring teratogenicity of C4F7N, an environmentally friendly insulating gas to replace SF₆. ResearchGate. https://www.researchgate.net/publication/366707605_Study_on_subacute_inhalation_toxicity_and_offspring_teratogenicity_of_C4F7N_An_environmentally_friendly_insulating_gas_to_replace_SF6

197. Toropova, A. P., Toropov, A. A., Roncaglioni, A., & Benfenati, E. (2023). The enhancement scheme for the predictive ability of QSAR: A case of mutagenicity. *Toxicology in Vitro*, 91, 105629. <https://doi.org/10.1016/j.tiv.2023.105629>
198. Trott, O., & Olson, A. J. (2010). AutoDock Vina: improving the speed and accuracy of docking with a new scoring function, efficient optimization, and multithreading. *Journal of computational chemistry*, 31(2), 455-461.
199. Tuccinardi, T. (2021). What is the current value of MM/PBSA and MM/GBSA methods in drug discovery? *Expert Opinion on Drug Discovery*, 16(11), 1233–1241. <https://doi.org/10.1080/17460441.2021.1942836>
200. Turkington, R. E., Hukriede, N. A., Ho, J., Jayasundara, N., & Sanders, A. P. (2025). Metal mechanisms of mitochondrial toxicity: recent review of arsenic, cadmium, and lead-induced nephrotoxicity. *Environmental science and pollution research international*, 32(24), 14439–14451. <https://doi.org/10.1007/s11356-025-36538-6>
201. Turlej, E. et al. (2024). Antimicrobial resistance patterns of *Streptococcus pneumoniae* isolates: A cross-sectional study. *Medicina*, 59 (11), 1927. <https://doi.org/10.3390/medicina59111927>
202. UniProt Consortium. (2021). UniProt: The universal protein knowledgebase in 2021 . *Nucleic Acids Research*, 49 (D1), D480–D489. <https://doi.org/10.1093/nar/gkaa1100>
203. Urs, A.R., Ganesan Selvi, V., Sreekumar, A. *et al.* Benchmarking web-based in-silico toxicity prediction tools using gold-standard datasets across five key endpoints. *Discov Appl Sci* **8**, 323 (2026). <https://doi.org/10.1007/s42452-026-08308-7>
204. Usman, M., Li, A., Wu, D., Qinyan, Y., Yi, L. X., He, G., & Lu, H. (2023). The functional role of lncRNAs as ceRNAs in both ovarian processes and associated diseases. *Non-coding RNA Research*, 9(1), 165–177. <https://doi.org/10.1016/j.ncrna.2023.11.008>
205. Utgés, J.S., Barton, G.J. Comparative evaluation of methods for the prediction of protein–ligand binding sites. *J Cheminform* **16**, 126 (2024). <https://doi.org/10.1186/s13321-024-00923-z>
206. Varadi, M. et al. “AlphaFold Protein Structure Database in 2024: Providing structure coverage for over 214 million protein sequences.” *Nucleic Acids Research*, gkad1011
207. Varadi, M. et al. “AlphaFold Protein Structure Database: massively expanding the structural coverage of protein-sequence space with high-accuracy models.” *Nucleic Acids Research*, 50(D1), pages D439–D444 (2021). DOI: 10.1093/nar/gkab1061
208. Vidal-Limon, A., Gonzalez-Perez, M., & Medina-Franco, J. L. (2022). Integration of molecular docking analysis and molecular dynamics simulations for studying food proteins and

- bioactive peptides. *Journal of Agricultural and Food Chemistry*, 70(3), 755–767. <https://doi.org/10.1021/acs.jafc.1c06110>
209. Walters, W. P., & Lazo, J. S. (2024). Structure-based drug design: Quantitative chemical insights. *Journal of Chemical Information and Modeling*. <https://doi.org/10.1021/acs.jcim.4c02122>
210. Wang, E., Liu, H., Wang, J., Weng, G., Sun, H., Wang, Z., Kang, Y., & Hou, T. (2020). Development and evaluation of MM/GBSA based on a variable dielectric GB model for predicting Protein–Ligand binding affinities. *Journal of Chemical Information and Modeling*, 60(11), 5353–5365. <https://doi.org/10.1021/acs.jcim.0c00024>
211. Wang, L., Laird-Fick, H. S., Parker, C. J., & Solomon, D. (2021). Using Markov chain model to evaluate medical students' trajectory on progress tests and predict USMLE step 1 scores---a retrospective cohort study in one medical school. *BMC medical education*, 21(1), 200. <https://doi.org/10.1186/s12909-021-02633-8>
212. Wang, N., Wang, X., Xiong, G., Yang, Z., Lu, A., Chen, X., Liu, S., Hou, T., & Cao, D. (2022). Machine learning to predict metabolic drug interactions related to cytochrome P450 isozymes. *Journal of Cheminformatics*, 14(1), 23. <https://doi.org/10.1186/s13321-022-00602-x>
213. Weber, J., et al. (2024). New perspectives on molecular epidemiology of antimicrobial resistance. *Microbial Spectrum*. <https://doi.org/10.1128/spectrum.03151-24>
214. Weng, G., Wang, E., Chen, F., Sun, H., Wang, Z., Hou, T., & Cao, D. (2024). Assessing the performance of MM/PBSA and MM/GBSA methods: Prediction reliability for protein–peptide complexes. *Physical Chemistry Chemical Physics*, 26(2), 1234–1245. <https://doi.org/10.1039/D3CP04366E>
215. Wills, S., Sanchez-Garcia, R., Dudgeon, T., Roughley, S. D., Merritt, A., Hubbard, R. E., ... Deane, C. M. (2023). Fragment merging using a graph database. *Journal of Chemical Information and Modeling*, 63(11), 3423–3437. <https://doi.org/10.1021/acs.jcim.3c00276>
216. Wolden, R., Pain, M., Karlsson, R., Karlsson, A., Aarag Fredheim, E. G., & Cavanagh, J. P. (2020). Identification of surface proteins in a clinical *Staphylococcus haemolyticus* isolate by bacterial surface shaving. *BMC Microbiology*, 20(1). <https://doi.org/10.1186/s12866-020-01778-8>
217. Wu, Y., Ma, L., Li, X., Yang, J., Rao, X., Hu, Y., Xi, J., ... Liu, S. (2024). AI in drug screening, design, and trials. *Frontiers in Pharmacology*, 15, 1459954. <https://doi.org/10.3389/fphar.2024.1459954>
218. Xiao Wang, Jia Wang, Han Zhang, Shenwei Huang, Yanbin Yin, HDMC: a novel deep learning-based framework for removing batch effects in single-cell RNA-seq data,

Bioinformatics, Volume 38, Issue 5, March 2022, Pages 1295–1303, <https://doi.org/10.1093/bioinformatics/btab821>

219. Yang Liu, Hanlin Li, Tianyu Zeng, Yang Wang, Hongqi Zhang, Ying Wan, Zheng Shi, Renzhi Cao, Hua Tang, Integrated bulk and single-cell transcriptomes reveal pyroptotic signature in prognosis and therapeutic options of hepatocellular carcinoma by combining deep learning, Briefings in Bioinformatics, Volume 25, Issue 1, January 2024, bbad487, <https://doi.org/10.1093/bib/bbad487>
220. Yang, M., Bo, Z., Xu, T., Xu, B., Wang, D., & Zheng, H. (2023). UniGBSA: An opensource and webbased automatic workflow to perform MM/GB(PB)SA calculations for virtual screening. Briefings in Bioinformatics, 24(4), bbad218. <https://doi.org/10.1093/bib/bbad218>
221. Yau, M. Q., Liew, C. W. Y., Toh, J. H., & Loo, J. S. E. (2024). A head-to-head comparison of MM/PBSA and MM/GBSA in predicting binding affinities for the CB₁ cannabinoid ligands. *Journal of molecular modeling*, 30(11), 390. <https://doi.org/10.1007/s00894-024-06189-4>
222. Ye, et al. (2024). CASTpFold: Computed Atlas of Surface Topography of the universe of protein Folds. <https://academic.oup.com/nar/article/52/W1/W194/7680624>
223. Yi, Q., Zhang, G., Wang, T., Li, J., Kang, W., Zhang, J., Liu, Y., & Xu, Y. (2025). Comparative Analysis of Metagenomic Next-Generation Sequencing, Sanger Sequencing, and Conventional Culture for Detecting Common Pathogens Causing Lower Respiratory Tract Infections in Clinical Samples. *Microorganisms*, 13(3), 682–682. <https://doi.org/10.3390/microorganisms13030682>
224. Yook, J., Jang, W., & Shin, W. (2025). AI-driven progress in fragment-based drug discovery. *Current Opinion in Structural Biology*, 91, 102995. <https://doi.org/10.1016/j.sbi.2025.102995>
225. York, et al. (2023). Modern Alchemical Free Energy Methods for Drug Discovery Explained. <https://pubs.acs.org/doi/10.1021/acsphyschemau.3c00033>
226. Zavitsanou, S., Tsengenes, A., Papadourakis, M., Amendola, G., Chatzigoulas, A., Dellis, D., Cosconati, S., & Cournia, Z. (2021). FEPprepare: A Web-Based Tool for Automating the Setup of Relative Binding Free Energy Calculations. *Journal of Chemical Information and Modeling*, 61(9), 4131–4138.
227. Zhang et al. (2024) MGMT activated by Wnt pathway promotes cisplatin tolerance through inducing slow-cycling cells and nonhomologous end joining in colorectal cancer <https://doi.org/10.1016/j.jpha.2024.02.004>
228. Zhang, H., Kim, S., & Im, W. (2022). Practical Guidance for Consensus Scoring and Force Field Selection in Protein-Ligand Binding Free Energy Simulations. *Journal of chemical information and modeling*, 62(23), 6084–6093. <https://doi.org/10.1021/acs.jcim.2c01115>

229. Zhang, J., & Li, L. (2022). Structural insights into the human genome and regulatory networks. *Nucleic Acids Research*, 50(1), 1–12. <https://doi.org/10.1093/nar/gkac1003>
230. Zhang, L., Rao, V., & Cornell, W. (2023). R-BRICS – a revised BRICS module that breaks ring structures and carbon chains. *ChemMedChem*, 19(4), e202300202. <https://doi.org/10.1002/cmdc.202300202>
231. Zhang, S. et al (2025) The role and application of bioinformatics techniques and tools in drug discovery <https://pmc.ncbi.nlm.nih.gov/articles/PMC11865229/>
232. Zhang, X. et al. (2023). In vitro antibacterial activities and mechanism of action of a novel synthetic peptide against multidrug-resistant bacteria. *Antibiotics*, 12 (2), 315. <https://doi.org/10.3390/antibiotics12020315>
233. Zhang, Y., Liu, H., & Chen, X. (2025). Molecular docking, free energy calculations, and ADMETox studies of chromene glycosides. *Frontiers in Chemistry*, 13, 1531152. <https://www.frontiersin.org/articles/10.3389/fchem.2025.1531152/full>
234. Zhang, Y., Zhang, Z., Ke, D., Pan, X., Wang, X., Xiao, X., & Ji, C. (2024). FragGrow: A web server for structurebased drug design by fragment growing within constraints. *Journal of Chemical Information and Modeling*, 64(10), 3970–3976. <https://doi.org/10.1021/acs.jcim.4c00154>
235. Zhao, et al. (2024). *Computational Systems Biology*. <https://onlinelibrary.wiley.com/doi/toc/10.1155/8086.si.905281>
236. Zhao, H. (2024). The science and art of structure-based virtual screening. *ACS Medicinal Chemistry Letters*, 15 (4), 436–440. <https://doi.org/10.1021/acsmedchemlett.4c00093>
237. Zhao, H. (2025). Rethinking Ligand Efficiency: Normalization pitfalls, uncertainty, and State-Invariant Metrics. *ACS Medicinal Chemistry Letters*. <https://doi.org/10.1021/acsmedchemlett.5c00652>
238. Zhao, Y., Cui, M., Lyu, Y., Gao, W., Cui, X., Zhang, C., & Meng, Q. (2023). External electric field effects on AFM-like magnetism in Nitroxide-Functionalized C+:C base pair. *Journal of Chemical Information and Modeling*, 63(7), 2037–2046. <https://doi.org/10.1021/acs.jcim.2c01489>
239. Zhou et al. (2024) Advances in machine-learning approaches to RNA-targeted drug design <https://www.sciencedirect.com/science/article/pii/S2949747724000113>
240. Zhou, et al. (2020). Recent advances in the development of protein–protein interactions modulators: mechanisms and clinical trials. <https://www.nature.com/articles/s41392-020-00315-3>
241. Zhou, M., Wang, Z., Zhang, L., Kudinha, T., An, H., Qian, C., ... Zhang, J. (2022). Serotype distribution and virulence in China. *Frontiers in Microbiology*, 12, 798750. <https://doi.org/10.3389/fmicb.2021.798750>

242. Zintgraff, J., Gagetti, P., Napoli, D., Eluchans, N. S., Irazu, L., Moscoloni, M., ... Corso, A. (2021). Invasive *S. pneumoniae* in Argentina. *Vaccine*, 40(3), 459–470. <https://doi.org/10.1016/j.vaccine.2021.12.008>
243. Zonyfar, C., Njimbouom, S. N., Mosalla, S., & Kim, J. (2024). GTransCYPs: an improved graph transformer neural network with attention pooling for reliably predicting CYP450 inhibitors. *Journal of Cheminformatics*, 16(1), 119. <https://doi.org/10.1186/s13321-024-00915-z>

Appendices

APPENDIX A

Letter of Approval from RDIC

 OUR LADY OF FATIMA UNIVERSITY INSTITUTIONAL ETHICS REVIEW COMMITTEE	R.I.S.E. Tower, Fatima Avenue Marulas, Valenzuela City 283 9754 loc.1204 ierc@fatima.edu.ph
February 23, 2026	
LI REN S. CHEN College of Medical Laboratory Science Our Lady of Fatima University Reference Number: <u>2026-IERC2-20171</u>	
Dear Mr. / Ms. Chen:	
We wish to inform you that the Our Lady of Fatima University Institutional Ethics Review Committee (OLFU-IERC) reviewed your study protocol entitled, AN IN-SILICO AND REVERSE FRAGMENT-BASED DRUG DESIGN VIA SMALL FRAGMENT LIBRARY AGAINST THE SELECTIVE PORTION OF THE PROTEIN-PROTEIN INTERACTOME NETWORK OF STREPTOCOCCUS PNEUMONIAE SEROTYPE 19A GUIDED BY FREE ENERGY PERTURBATION (FEP) SIMULATIONS on February 23, 2026, upon review of study protocol, the panel action is EXEMPTION FROM ETHICAL REVIEW.	
JUSTIFICATION FOR THIS DECISION:	
<u>FOR RESEARCHES WITH NO STUDY PROTOCOL AND DOES NOT CONSITUTE HUMAN HEALTH RESEARCH:</u>	
This activity does not constitute human health research and will not involve collection of individual identifiable data. Furthermore, the National Ethical Guidelines for Research Involving Human Participants (2022) states that research that only includes interactions involving survey procedures, interview procedures, or observation of public behavior, "if the information obtained is recorded by the investigator in such a manner that the identity of the participant cannot be ascertained", is considered exempt from ethical review.	
<u>Document/s included in the review on which this exemption was based:</u>	
<u>Ethics Application Form, Version 1, February 3, 2026.</u> <u>Study Protocol and Informed Consent Checklist, Version 1, February 3, 2026.</u> <u>Research Manuscript, Version 1, February 3, 2026.</u> <u>Curriculum Vitae, Version 1, February 3, 2026.</u> <u>Endorsement Letter, Version 1, February 23, 2026.</u>	
RESPONSIBILITIES OF THE PRINCIPAL INVESTIGATOR/RESEARCHER WHILE THE STUDY IS IN PROGRESS AND UPON COMPLETION:	
1. Continuing compliance with the exemption criteria of the National Ethical Guidelines for Research Involving Human Participants (2022) in the duration of the study; 2. No substantial changes shall be made in the research design, methodology, and subject population from the protocol submitted for exemption. Changes that will have an impact on the risk-benefit assessment or qualification for exemption may be submitted as a new protocol for initial review. If there is a need to amend the protocol	



R.I.S.E. Tower, Fatima Avenue
Marulas, Valenzuela City
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ierc@fatima.edu.ph

- or ICF, researcher should submit an amendment review request (using FORM10(a) Protocol Amendment Form).
3. The principal investigator submits an accomplished FINAL REPORT FROM (FORM14(a) Final Report Form) at the completion of the study.

All further queries regarding this decision may be forwarded to the undersigned through ierc@fatima.edu.ph.

Thank you.

Yours,

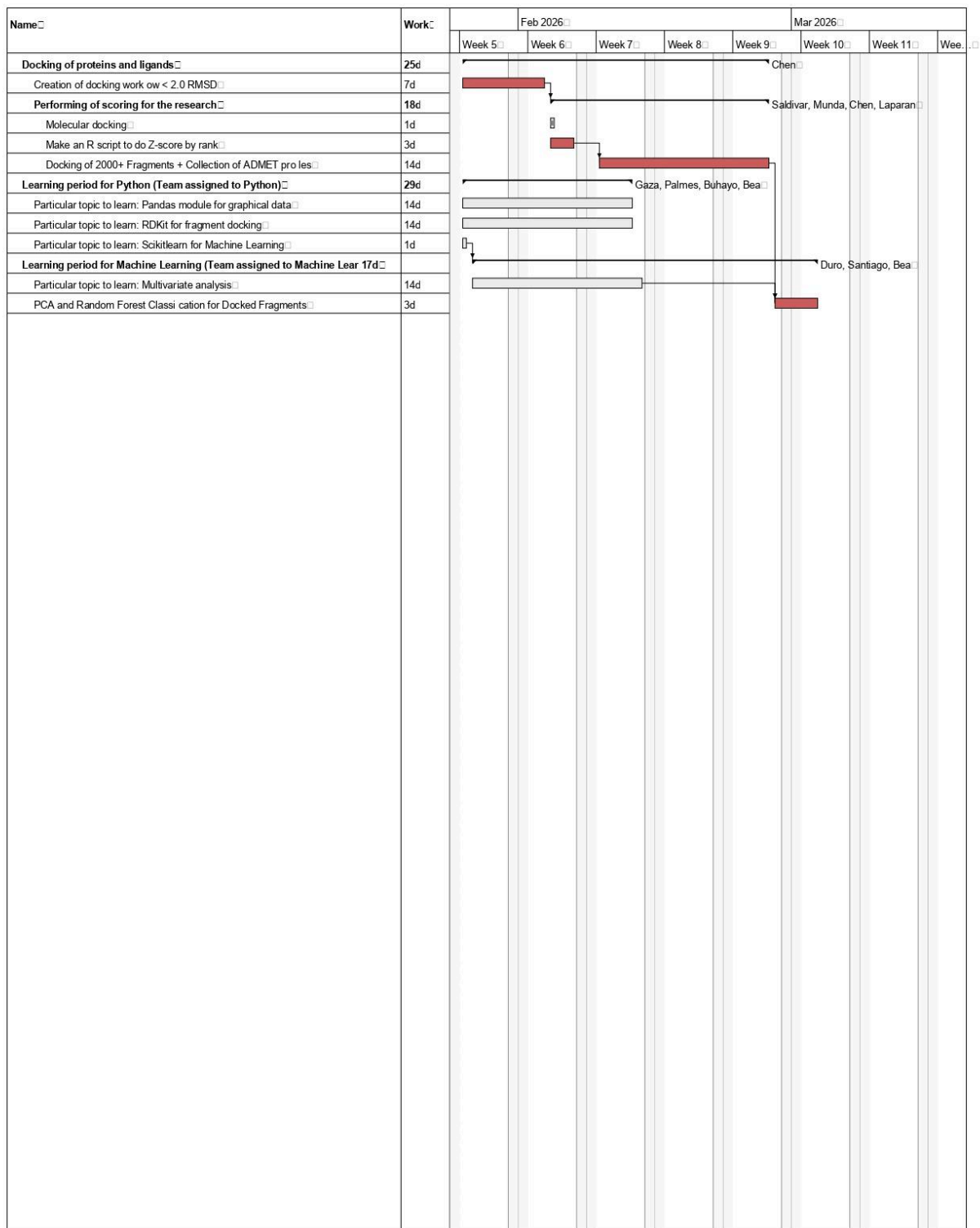
Jenica Ana Rivero, MAN, RN, PGDipHS
Chair, Institutional Ethics Review Committee
Our Lady of Fatima University

APPENDIX B
Research Budget

Item	Cost (₱)
Research Manuscript Fineprint	3,500
Updated E-Journals	2,500
Acer Aspire A715-59G	47,999
Research Validation	3,000
Coding Courses	2,500
Total	50,499

APPENDIX C

Timeline



APPENDIX D

Certification

CERTIFICATE OF VALIDATION

This is to certify that the study entitled,

In-silico and reverse fragment-based drug design via small fragment library against the selective portion of the protein-protein interactome network of *Streptococcus pneumoniae* serotype 19A guided by free energy perturbation (FEP) simulations

prepared and authored by,

Li Ren S. Chen, Jon Rex G. Bea, Jan Merrik F. Buhayo, Aicy Marie D. Duro, Joshua Kim C. Gaza, Khate F. Laparan, Ma. Lirah Y. Munda, Krist Hayden B. Palmes, Sheryl C. Saldivar, Xyrill Jade D. Santiago, Mikhail Kristoffer S. Gonong, Ed Joshua Montera, John Kenneth Pagdanganan

has received interpretation and validation support.

The proper interpretation and validation methods were appropriately used and employed based on the results and currently applied design of the study.

This certification attests to the quality and accuracy of the interpretation and validation work completed.



TIMOTHY M. MENDOZA
Validator

Affiliation: University of the Philippines Diliman
E-mail address: tmthy.mln@gmail.com
Date: May 4, 2026

Date: May 4, 2026

Requesting Research Team: Our Lady of Fatima University - Chen, et. al.

Method: Fragment-based drug design (FBDD) against *S. pneumoniae* serotype 19A


Validated by: Timothy M. Mendoza
MS Candidate, University of the Philippines Diliman

EXECUTIVE SUMMARY

The study presents a comprehensive and well-structured computational pipeline for fragment-based drug discovery targeting *Streptococcus pneumoniae*. The integration of bioinformatics, structural biology, chemo-informatics, and computational chemistry demonstrates a multidisciplinary and systematic approach aligned with current practices in computer-aided drug design (CADD).

Overall, the workflow effectively progresses from target identification to candidate prioritization, employing multiple filtering and evaluation steps that collectively support the identification of promising ligand candidates.

SCOPE OF VALIDATION

This report evaluates the valid methodological components and analytical steps of the study, including:

- Target identification and prioritization
- Protein structure modeling and assessment
- Fragment library design and preparation
- Molecular docking and ligand evaluation
- Chemoinformatics-based clustering
- Reverse docking analysis
- Free Energy Perturbation (FEP) application
- ADMET prediction

METHODOLOGICAL STRENGTHS

1. Systematic Target Identification and Selection – using verified and curated databases ensures biologically meaningful target selection
2. Protein Structure Modeling and Quality Assessment – AlphaFold2 is the gold standard in structural prediction, considering confidence metrics such as pLDDT supports structure reliability assessment
3. Binding Site Identification and Druggability Analysis – using topology-based tools supports accurate localization of ligand-binding regions

4. Systems Biology Integration via PPI Networks – using STRING confidence scores introduces a quantitative measure of interaction reliability
5. Fragment Library Design Using Rule of Three (Ro3) – alignment with established FBDD principles supports efficient chemical space exploration
6. Ligand Preparation Using Standard Chemoinformatics Tools – the use of widely accepted tools ensures reproducibility and standardization and energy minimization improves structural realism of ligands prior to docking
7. Molecular Docking for Primary Screening - AutoDock Vina is a well-established docking engine and provides an efficient method for initial screening of large fragment libraries
8. Ligand Efficiency-Based Prioritization – LE is a recognized metric in fragment-based drug design, enabling identification of highly efficient small molecules
9. Chemoinformatics-Based Clustering (Tanimoto + Butina) - these are standard chemoinformatics techniques that reduces redundancy of candidates and supports diversity-oriented selection of fragments
10. Reverse Docking for Target Interaction Profiling – provided additional insight into ligand selectivity and promiscuity
11. Application of Free Energy Perturbation (FEP) – FEP is considered a high-level computational method for estimating binding free energies, its inclusion demonstrates an advanced attempt at quantitative validation
12. ADMET Prediction for Drug-Likeness Assessment – assists in prioritizing candidates with favorable pharmacokinetic profiles, the use of valid and widely used databases ensures well-informed selection
13. Multi-Criteria Decision-Making for Final Candidate Selection - the study integrates Docking results, Ligand efficiency, Clustering, FEP outputs, and ADMET predictions which reflects real-world drug discovery strategies and enhances robustness of candidate prioritization

The study successfully implements a comprehensive and methodologically sound computational pipeline for fragment-based drug discovery. The integration of multiple computational techniques—from bioinformatics-driven target selection to chemoinformatics filtering and advanced energy calculations—demonstrates a well-rounded and scientifically grounded approach.

Overall, the study represents a strong example of an end-to-end in silico drug discovery framework, suitable for generating testable hypotheses and guiding future experimental validation.

RECOMMENDATION

- Perform replicates of docking simulations and implement statistical tests to increase robustness of computations
- Mention that high centrality in PPI networks does not automatically translate to druggability. Human homologs can also be filtered out in the selection.
- Once resources become available, the fragment library can be further expanded to reduce sampling bias and increase diversity (reduces false negatives)

APPENDIX E

Terminologies

This appendix provides simplified definitions of key scientific and medical terms used throughout the study, serving as a quick reference guide for readers.

Antibiotics – Medicines that kill bacteria.

Drugs – Substances used to treat diseases.

Combinatorial drugs – Using two or more medicines together.

PCV – Vaccine against *Streptococcus pneumoniae*.

NVT – Types of *Streptococcus pneumoniae* not covered by the vaccine.

AMR – Germs resisting medicines.

Streptococcus pneumoniae – Bacterium causing pneumonia and other infections.

PBP – Bacterial proteins that block penicillin.

Gene – DNA instructions for proteins.

Genetic changes – Alterations in DNA.

Gene annotation – Labeling parts of DNA.

Gene assembly – Piecing together full genes.

Bioinformatics – Using computers to study biological data.

CGA – Genome analysis service.

FASTA – Text format for DNA/protein sequences.

In silico – Computer-based experiments.

Molecular docking – Modeling drug-target fit.

Docking score – Number predicting binding strength.

Binding affinity – Strength of drug-target bond.

FEP – Method to calculate binding energy changes.

LE – Efficiency of a molecule compared to its size.

Fragment – Small molecule used in drug design.

Alchemy – Predictive chemistry.

SAR – Link between structure and activity.

PPI – Studying protein interactions.

Protein – Large molecules made of amino acid.

NMR – Technique to study protein structures.

APPENDIX F

About the Researchers



CURRICULUM VITAE

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Marital Status	: Single
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Permanent Residence	: Philippines

II. Educational Background

PRIMARY:

Calauag Elementary School

SECONDARY:

Junior Highschool	: Naga College Foundation inc.
Senior Highschool	: Naga College Foundation inc.

TERTIARY

Our Lady of Fatima University, Quezon City

III. Affiliation/Organization

College of Medical Laboratory Science
Our Lady of Fatima University

IV. Achievements

Senior Highschool : With high honor (Gr 12)
Junior Highschool : With honor (Gr 11)



OUR LADY OF FATIMA UNIVERSITY



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PRIMARY:

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Senior Highschool	: Our Lady of Fatima University (STEM)

TERTIARY:

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III. Affiliation/Organization

College of Medical Laboratory Science
Our Lady of Fatima University

IV. Achievements



OUR LADY OF FATIMA
UNIVERSITY



Elementary School:

- Mr. United Nations 2012 - 1st Place (S.Y. 2012-2013)
- QCC-BSP Kab Scout Palaro 2013 - Best in Headgear (S.Y. 2013-2014)
- Nutrition Month 2013 - Best in Costume (S.Y. 2013-2014)
- Ginoong/Binibining Kalikasan 2014 - 2nd Place (S.Y. 2014-2015)
- BIO NEP 2014 - 2nd Place (S.Y. 2014-2015)
- Mr. United Nations 2014 - 1st Place (S.Y. 2014-2015)
- Nutrition Month 2016 Cooking Contest - 1st Place (S.Y. 2016-2017)

Jr. High School:

- Certificate of Participation: Math and ICT Summer Camp (S.Y. 2017-2018)

Senior High School:

- Endemic FaunArt Poster Making - 2nd Place (S.Y. 2021-2022)

Undergraduate Studies:

- Dean's Lister (S.Y. 2024-2025)
- Certificate of Participation: Three Day Genomics and Bioinformatics Workshop in GoGeneBio (S.Y. 2025-2026)



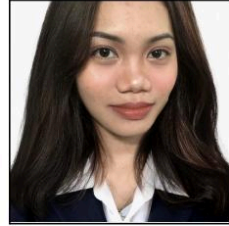
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Permanent Residence	: Philippines

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Senior Highschool : Paradise Farm National Highschool

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College of Medical Laboratory Science of Our Lady of Fatima University,
Quezon City Campus

IV. Achievements

With Honor (S.Y 2016-2017)

Academic Distinction (2017-2019)

With Honors (S.Y 2020-2021)

With High Honors (S.Y 2021-2022)

With Honors (S.Y 2022-2023)



OUR LADY OF FATIMA UNIVERSITY



CURRICULUM VITAE

JOSHUA KIM C. GAZA

14 Senator St, Sta. Monica Village, Barangay Gulod, Quezon City

jcgaza0257qc@student.fatima.edu.ph

09934086112



I. Personal Information

Date of birth : October 20, 2004
Age : 21
Sex : M
Marital status : Single
Nationality : Filipino
Permanent Residence : Philippines

II. Educational Background

PRIMARY:

John Mark Christian School

SECONDARY:

Junior Highschool : San Bartolome High School
Senior Highschool : STI Novaliches

TERTIARY:

Bachelor : Our Lady of Fatima University (Medical Technology)

III. Affiliation/Organization

College of Medical Laboratory Science of Our Lady of Fatima University,
Quezon City Campus

IV. Achievements

Senior High School : With Honors (Grade 11-12)
Best Use of SHS iLS Expo Theme
Most Promising Company (SHS iLS Expo)
Member, Alab Lahi Club (Senior High School)

Junior High School : With Honors

Elementary : Valedictorian
Subject Excellence Awards



CURRICULUM VITAE

KHATE FERNANDEZ LAPARAN

Blk. 64 purok I, Brgy. sto Nino II, San Jose Del Monte Bulacan

kflaparan0162qc@student.fatima.edu.ph

0945-367-8963



I. Personal Information

Date of Birth : December 16, 2004
Age : 21
Sex : Female
Marital Status : Single
Nationality : Filipino
Permanent Residence : Philippines

II. Educational Background

PRIMARY:

Liceo De Daniel School Inc.
School of St. Ignatius Inc.

SECONDARY:

Junior Highschool : School of St. Ignatius Inc.
Senior Highschool : (STEM Strand) Our Lady of Fatima University, Quezon City

TERTIARY:

Our Lady of Fatima University, Quezon City

III. Affiliation/Organization

College of Medical Laboratory Science of Our Lady of Fatima University,
Quezon City Campus

IV. Achievements

Higher Academic Distinction (S.Y. 2018-2019)
General Honor Award (S.Y. 2019-2020)
General Honor Award (S.Y. 2020-2021)
General Honor Award (S.Y. 2022-2023)
Subject Excellence in Pre-Calculus & Physics 1 (S.Y 2021-2023)



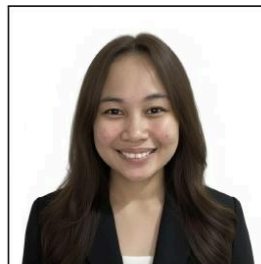
CURRICULUM VITAE

MA. LIRAH Y. MUNDA

372 AREA 9, DURIAN ST., BRGY. PASONG TAMO, QUEZON CITY

mymunda0077qc@student.fatima.edu.ph

09089281869



I. Personal Information

Date of birth : July 19, 2005
Age : 20
Sex : F
Marital status : Single
Nationality : Filipino
Permanent Residence : Philippines

II. Educational Background

PRIMARY:

Tumpa Elementary School

SECONDARY:

Junior Highschool : Marcial O. Rañola Memorial School
Senior Highschool : (STEM Strand) Marcial O. Rañola Memorial School

TERTIARY:

Our Lady of Fatima University, Quezon City

III. Affiliation/Organization

College of Medical Laboratory Science of Our Lady of Fatima University,
Quezon City Campus

IV. Achievements

CHED Scholar (2023 - Present)
With Honor (S.Y. 2022-2023)
With High Honor (S.Y. 2021-2022)
With Honor (S.Y. 2020-2021)
With Honor (S.Y. 2019-2020)



CURRICULUM VITAE

KRIST HAYDEN B. PALMES

CamiaSt,Brgy SanMartinII, SanJose Del Monte, Bulacan
kristhaydenpalmes@gmail.com
+639284218659



I. Personal Information

Date of birth	: August 1, 2005
Age	: 20
Sex	: M
Marital status	: Single
Nationality	: Filipino
Permanent Residence	: Philippines

II. Educational Background

PRIMARY:

Emilio Jacinto Elementary School
Kalawaan Elementary School

SECONDARY:

Junior Highschool	: San Joaquin-Kalawaan High School
Senior Highschool	: San Jose Del Monte National High School

TERTIARY:

Bachelor	: Our Lady of Fatima University (Medical Technology)
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III. Affiliation/Organization

College of Medical Laboratory Science - Our Lady Of Fatima University

IV. Achievements

With Honors - Senior High School



CURRICULUM VITAE

SHERYL CABADING SALDIVAR

Blk 5 Lot 15 Mirandaville Tawid Sapa 2, Brgy., Kaligayahan Nova Q.C.
scabading3943qc@student.fatima.edu.ph
09922474465



I. Personal Information

Date of birth	: July 7, 2004
Age	: 21
Sex	: Female
Marital status	: Single
Nationality	: Filipino
Permanent Residence	: Philippines

II. Educational Background

PRIMARY:

San Agustin Elementary School

SECONDARY:

Junior Highschool : Novaliches High School

Senior Highschool : Our Lady of Fatima University, Quezon City Campus

TERTIARY:

Our Lady of Fatima University

III. Affiliation/Organization

College of Medical Laboratory Science of Our Lady of Fatima University,
Quezon City Campus

IV. Achievements

Academic Awardee (S.Y 2015-2016)

With Honor (S.Y. 2017-2018)

With High Honor (S.Y 2019-2020)

Subject Excellence Award (S.Y 2021-2022)

With High Honor (S.Y 2022-2023)



CURRICULUM VITAE

XYRILL JADE D. SANTIAGO

GREATER LAGRO ESPERANZA SUBDIVISION, QUEZON CITY

xdsantiago3547qc@student.fatima.edu.ph

0945-723-8998



I. Personal Information

Date of birth	: May 14, 2004
Age	: 21
Sex	: Male
Marital status	: Single
Nationality	: Filipino
Permanent Residence	: Philippines

II. Educational Background

PRIMARY:

Buenavista Elementary School

SECONDARY:

Junior Highschool	: Sablayan National Comprehensive High School
Senior Highschool	: Sablayan National Comprehensive High School

TERTIARY:

Our Lady of Fatima University, Quezon City

III. Affiliation/Organization

College of Medical Laboratory Science
Our Lady of Fatima University - Quezon City

IV. Achievements

With High Honor (S.Y. 2022-2023)
With High Honor (S.Y. 2021-2022)
With High Honor (S.Y. 2020-2021)